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THE PALAEOMAGNETISM OF 2400 M.Y. OLD ROCKS FROM THE AUSTRALIAN PILBARA CRATON AND ITS RELATION TO ARCHAEAN-PROTEROZOIC TECTONICS

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ABSTRACT

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Palaeomagnetic results are described from 16 sites distributed in two traverses across the Black Range dyke. The dyke, a quartz gabbro dated radiogenically at 2329 ± 89 m.y., is one of the principal members of the NNE trending Black Range Suite which intruded the Archaean assemblage of granite plutons and greenstone belts of the Pilbara craton at around the time of the Archaean-Lower Proterozoic boundary. Results are also presented from 9 sites in the Cajuput dyke. The field evidence is consistent with this dyke having acted as a feeder for the overlying Mount Roe Basalt, the basal member of the Lower Proterozoic Fortescue Group. The Cajuput dyke also trends NNE and is thought to belong to the Black Range Suite. Samples of metamorphosed granite collected in the baked contact zone of the Black Range dyke exhibit various degrees of remagnetization. Directions of magnetization in material from the remelted granite margins are close to the mean direction of magnetization for the dyke. Beyond about 30 m from the contact, the effects of remagnetization are absent.

The pole positions computed from site-mean virtual geomagnetic poles for the Black Range dyke and the Cajuput dyke lie at lat. 32° S, long. 154° E (A_{95} = 9°) and lat. 46° S, long. 146° E (A_{95} = 22°), respectively. They fall on the Australian apparent polar wander curve drawn for the period pre-2000 m.y. which until now was based primarily on data from the Yilgarn craton. These data suggest that it is improbable that the younger, intervening mobile belt between the Yilgarn and Pilbara cratons was generated through the process of a plate tectonic, continent/continent-type convergence.

INTRODUCTION

Interest in Precambrian palaeomagnetism has been stimulated by current views on crustal dynamics and, in particular, by the question of whether the plate tectonic hypothesis is a valid explanation of the occurrence of certain tectonic features common to many shield areas. Amongst those features are mobile belts, greenstone belts and ancient fracture zones which commonly separate a mosaic of cratons, themselves constituting a larger Precambrian

platform. The western two-thirds of the Australian continent approximates this notion of a first-order geological subdivision.

The role of palaeomagnetism in assessing the drift history of individual cratons ('palaeoblocks') was considered by Hospers and Van Andel in 1969. Since then, the data reviewed for the shield regions of North America (Irving and Park, 1972; McGlynn et al., 1975), Africa (Piper et al., 1973) and Australia (McElhinny and Embleton, 1976) confirm that respective single apparent polar-wander curves can be drawn to accommodate the pole positions without violating the chronological constraints available. However, it has also been pointed out by Piper et al. (1973) and McGlynn et al. (1975) that this is merely the simplest interpretation of the data. The opening and closing of small oceans of the order of 10^3 km would probably remain undetected within the resolution of the data. Relative motion will also be undetectable if the Eulerian pole, which describes the relative motion between two crustal units, is close to their palaeomagnetic pole position (i.e., the indeterminacy of longitude).

THE PROJECT

The main purpose of the present study is to compare palaeomagnetic data from two of Australia's principal cratons, the Yilgarn and the Pilbara, for the period about 2400 m.y. ago, which corresponds approximately to the Archaean-Proterozoic boundary. Many results for the Australian Lower Proterozoic are now available (Irving and Green, 1958; Chamalaun and Porath, 1968; Evans, 1968; Porath and Chamalaun, 1968; Facer, 1971; Duff and Embleton, 1976; Giddings, 1976; Giddings and Embleton, 1976; McElhinny and Evans, 1976) and, in their review, McElhinny and Embleton (1976) have drawn attention to the 'single continent' model consistent with those data, particularly for Precambrian times younger than 1800 m.y. The sampling, however, has been biased towards the western part of the platform, rock units from the Yilgarn craton being particularly well represented. This craton, which comprises southwestern Australia, is separated from the Pilbara craton in northwestern Australia by a mobile belt, the Gascoyne Province and the Ophthalmian fold belt. The major movements in this zone took place after the time of the main granite emplacement in the adjacent cratons (Daniels, 1975). The geology of the western shield is summarized in Fig. 1. In that figure, a regional subdivision of the Precambrian rocks has been made following Trendall (1975). Although the two major cratons comprise essentially an association of granite plutons and greenstone belts, their tectonic styles are quite distinct: the Yilgarn craton displays a pronounced linear arrangement of greenstone belts whereas in the Pilbara craton, the granite plutons are encircled by narrow arcuate greenstone belts (see Fig. 1). Proposed correlations between the greenstones in the cratons have been discussed by Gee (1974) and more recently Rutland (1976) has reviewed the orogenic evolution of this shield region.

In the Yilgarn craton the approximate age range of the granite and gneiss

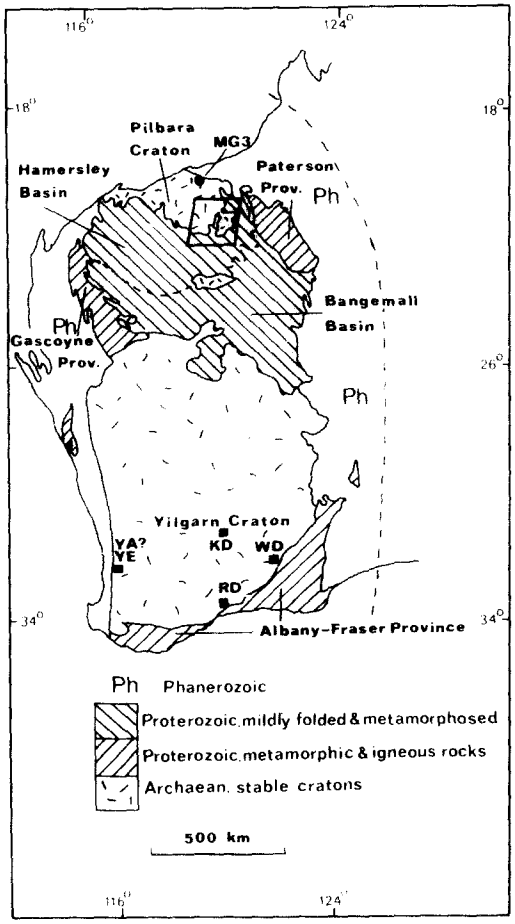
is 2900—2500 m.y. and in the mobile belt to the south, the Albany-Fraser Province, the age range is about 1300—1100 m.y. There are two main intervals of granite emplacement in the Pilbara craton, one around 3000 m.y. and a younger event when those granite batholiths were intruded by irregular masses of adamellite (the tin-bearing granites of Blockley, 1970) at around 2600 m.y. To the southwest and southeast respectively of the Pilbara craton, the Gascoyne and Paterson Provinces contain rocks formed by igneous and metamorphic activity with ages in the range 1800—1300 m.y., i.e., the interval approximately corresponding to time breaks in the deposition within the Hamersley and Bangemall Basins (Trendall, 1975).

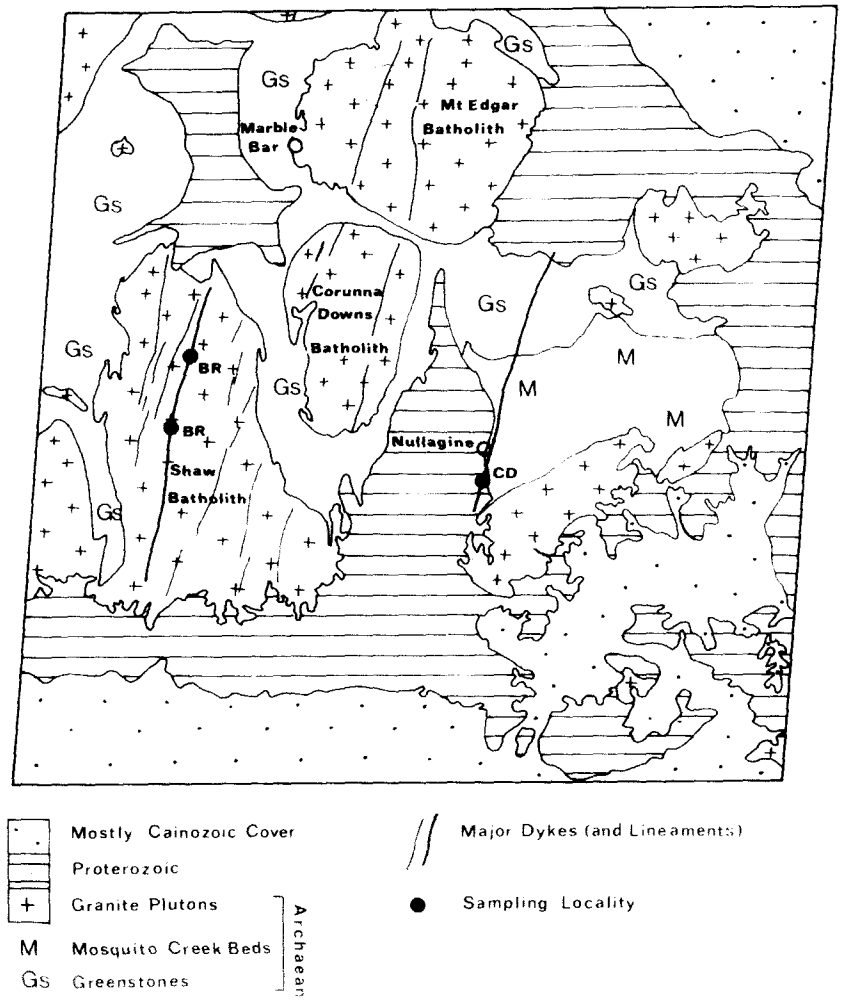
Compared to the Yilgarn craton, few palaeomagnetic data are available for the Pilbara craton. Porath and Chamalaun (1968) published results from the iron formations at Mount Goldsworthy, Mt. Tom Price and Mt. Newman but the assigned age constraints are broad and largely preclude the use of those data for detecting inter-cratonic movements. Lewis et al. (1974) have reported a Rb-Sr radiometric age of 2329 ± 89 m.y. for the Black Range dyke. Good age control is now available for the intrusions studied here (see below) enabling comparison with results from the Yilgarn craton (Turek, 1966; Evans, 1968; Giddings, 1976). The pole position for the Widgiemooltha Dyke Suite (dated at 2420 ± 30 m.y.) obtained by Evans (1968) was ranked as a key pole by McElhinny and Embleton (1976): their classification system was based equally on the precision of the palaeomagnetic and radiometric data.

GEOLOGICAL OUTLINE AND SAMPLING PROCEDURES

The sampling regions lie within the 1:250,000 sheets for 'Marble Bar' and 'Nullagine' (see Fig. 1). The regional geology of this area was first described by Noldart and Wyatt (1962) and more recently by Hickman and Lipple (1974) and Hickman (1975). At about 2400 m.y., the granite plutons of the Pilbara craton were host to a major intrusive phase known as the Black Range Suite. The Black Range dyke is one of the principal members of the NNE-trending suite that cuts the plutons and, according to Hickman and Lipple (1974), it has a total length of over 100 km and is up to 150 m wide. They have postulated that the dyke may have been a feeder for the Mount Roe Basalt, the basal member of the Lower Proterozoic Fortescue Group. The precise stratigraphic relationship of the Black Range dyke to the Archaean-Proterozoic unconformity and the rocks of Lower Proterozoic age is unknown. The Fortescue Group is preserved in the downthrown side of the fault which terminates the Black Range dyke to the north of the Shaw Batholith.

The Cajuput dyke (Hickman, 1975) is another major basic dyke parallel to, and presumably belonging to the same phase of activity as, the Black Range dyke, and about 50 km east of it. It has an outcrop width of up to a kilometre, and runs NNE for about 50 km from a point about 20 km south of Nullagine township. For this dyke there is some field evidence to relate the event to the Archaean-Proterozoic unconformity. It has been suggested by Hickman


116°
124°
18°
26°
34°
Pilbara
Craton
MG3
Hamersley
Basin
Paterson
Prov.
Ph
Bangemall
Basin
Gascoyne
Prov.
Yilgarn Craton
YA?
YE
KD
WD
RD
Albany–Fraser Province
Ph Phanerozoic
Proterozoic, mildly folded & metamorphosed
Proterozoic, metamorphic & igneous rocks
Archaean, stable cratons
500 km


Gs
Mt Edgar
Batholith
Marble
Bar
Corunna
Downs
Batholith
BR
BR
Shaw
Batholith
Nullagine
CD
M
Mostly Cainozoic Cover
Proterozoic
Granite Plutons
Mosquito Creek Beds
Greenstones
Archaean
Major Dykes (and Lineaments)
Sampling Locality

(1975) that in the northern part of its outcrop the Cajuput dyke may partly intrude the overlying Mount Roe Basalt. However, in the southern part the dyke is clearly overlain unconformably by the upper part of the Hardey Sandstone and by the Kylena Basalt, formations which successively overlie the Mount Roe Basalt. While it is not inconceivable that the Cajuput dyke acted primarily as a fissure type of feeder for the upper part of the Mount Roe Basalt, and was subsequently covered by higher formations after local erosion, it is also possible that the dyke is everywhere older than the Mount Roe Basalt (Trendall, 1976, personal communication).

The Black Range dyke was sampled at Cooglegong Creek (21.58° S, 119.47° E), viz., the section sampled for the geochronology. Fourteen oriented cores were taken with a hand-held power drill in the dyke and 5 oriented blocks were collected from the baked contact. 16 km south, a second traverse was made through the dyke in the Shaw River section at Hillside (21.72° S, 119.40° E). Here, 31 drill cores were collected from the dyke and 7 hand samples in the host granite. Good exposures of fresh dolerite were encountered at both of these localities: the superimposed drainage system has produced locally incised water courses traversing the dyke approximately normal to its strike direction. The present attitude of the dyke has been assumed to be close to its primary attitude since the simple linearity of the dyke's surface expression testifies to the absence of significant post-intrusion intra-batholith deformations. Systematic discrepancies between the directions of magnetization measured at the two sections point to minor deformation. Epeirogenic tilting is not excluded.

A single sampling traverse comprising 25 drill cores was made across the Cajuput intrusion approximately 7 km south of Nullagine township (21.95° S, 120.10° E).

Each sampling traverse involved drilling generally two or more cores at regular intervals across the intrusions wherever suitable outcrop permitted. Groups of cores collected in this way have been regarded as sites for the purpose of the analysis of the magnetic remanence directions. Whether the distribution of the sampling sites has covered a sufficiently long time span

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Fig. 1. The principal geological subdivisions of the western Precambrian shield following Trendall (1975). The right-hand map (side length approximately 185 km) shows some detail of the geology in the sampling area, outlined in the southeastern part of the Pilbara craton. The geology was interpreted from an overlay of the LANDSAT image, number 1148–01282–Band 5 for Marble Bar, northwestern Australia: this procedure is similar to that adopted by Viljoen et al. (1975). Digital image enhancement was carried out by the CSIRO Division of Mineral Physics.

The Black Range Dyke and the Cajuput dyke are indicated by the heavy trend lines; dykes and lineaments associated with the Black Range Suite are indicated by fine trend lines. Sampling sites in the Pilbara craton are denoted by large closed dots and sites from which results have been previously reported from the Yilgarn craton—see the Discussion—are denoted by square symbols. Letters designating all of the sites refer also to the palaeomagnetic pole positions plotted in Fig. 7.

to average Precambrian secular variation is a matter for conjecture. Sampling has covered 150 m in the Black Range dyke and approximately 300 m in the Cajuput dyke.

PALAEOMAGNETIC RESULTS

The Black Range dyke

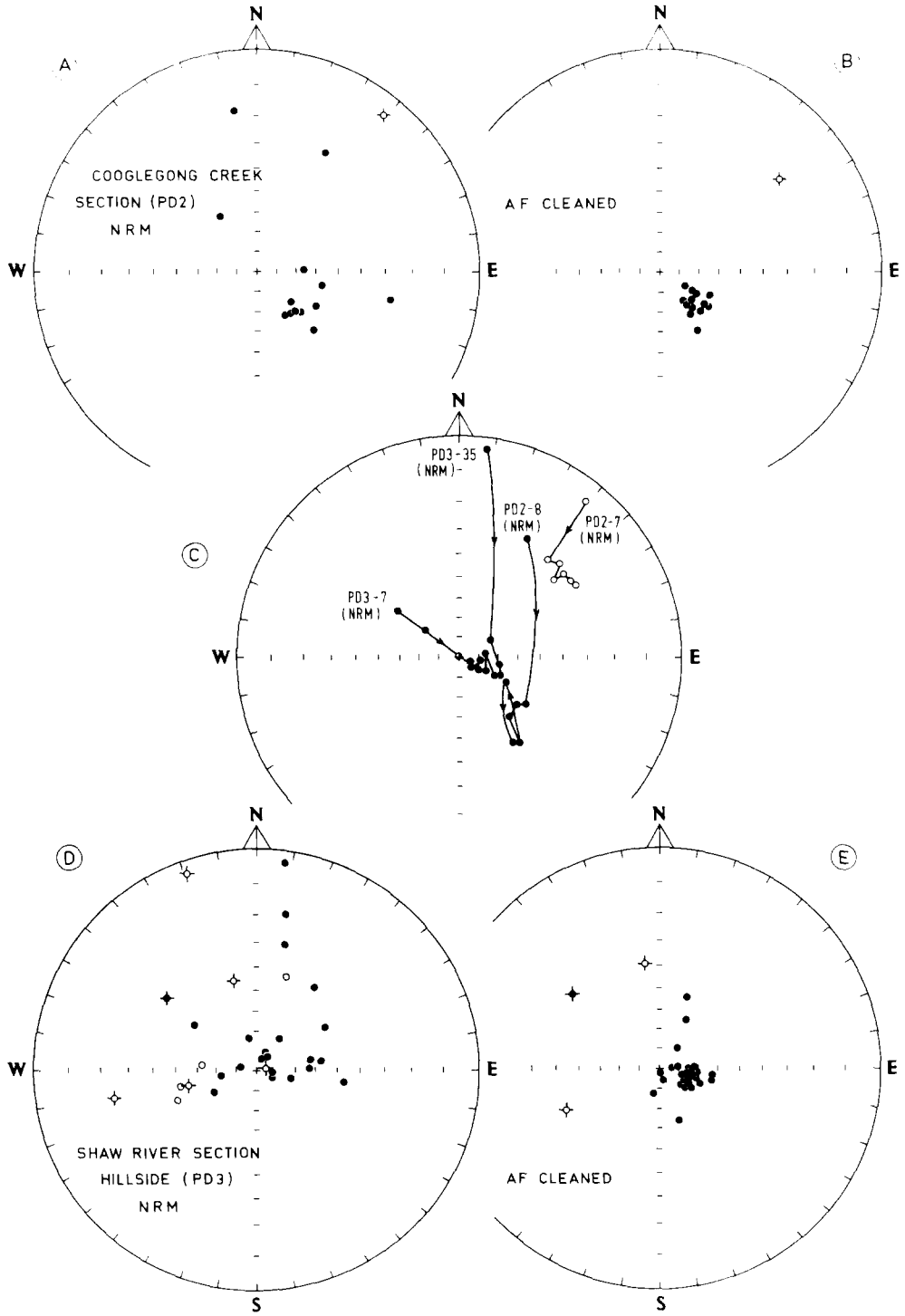
The natural remanent magnetization (NRM) directions were systematically distributed between a steep positive direction and a shallow north direction. The values for the intensity of NRM of samples from the section at Cooglegong Creek were systematically lower (average = 5 $mA.m^{-1}$) than the NRM intensities measured in the samples from Hillside (average = 203 $mA.m^{-1}$). The distribution of the NRM directions is shown in Fig. 2A, D.

One specimen from each core collected in the Cooglegong Creek section was partially demagnetized in alternating fields (AF) at 10 mT field increments to a maximum of 60 mT. The average demagnetization/intensity curve for the pilot specimens is shown in Fig. 3A. The shape of the curve is indicative of the presence of a large proportion of high coercive-force components. The minimum scatter was achieved after treatment at 20 mT (i.e. the precision parameter k (Fisher, 1953) reached a peak value). Two examples of the response of directions of magnetization to AF treatment are shown in Fig. 2C. Those specimen directions initially well grouped changed little upon cleaning whereas many of the directions initially oblique, rotated systematically to produce a tight cluster. After AF cleaning, only specimens from core PD2-7 remained oblique to a clearly well-defined group of stable directions. The coercive force spectrum of PD2-7 was also anomalous in that the median demagnetization field was < 10 mT and the normalized intensity was 0.09 after treatment at 60 mT (cf., the average curve in Fig. 3A). An optimum cleaning field of 20 mT was selected and all of the remaining specimens were treated accordingly. The 14 core-mean directions are shown in Fig. 2B. The variation of the declination and inclination in the intrusive with distance from the contacts is shown in Fig. 3B. Although there is no apparent systematic variation in the declination there is a suggestion of cyclicity in the inclination values.

Samples collected in the Shaw River section at Hillside were treated following similar procedures. The average demagnetization-intensity curve is shown in Fig. 4A. The minimum scatter for the pilot specimens was also achieved after cleaning at 20 mT AF. Two examples of direction changes

Fig. 2. Core-mean directions of magnetization for the Black Range dyke shown before and after AF treatment for the Cooglegong Creek (A and B) and the Shaw River (D and E) sections. Symbols marked with a cross denote those results which have been omitted from the analysis of directions (see text). Part C shows some examples of direction trends revealed during partial AF demagnetization at 10 mT field increments to 60 mT.

Closed (open) circles represent downward (upward) pointing directions. They are plotted on a Wulff equal-angle stereographic projection.


A
N
W
E
COOGLEGONG CREEK
SECTION (PD2)
NRM
B
N
E
AF CLEANED
C
N
W
E
PD3-35
(NRM)
PD2-8
(NRM)
PD2-7
(NRM)
PD3-7
(NRM)
D
N
W
E
S
SHAW RIVER SECTION
HILLSIDE (PD3)
NRM
E
N
E
S
AF CLEANED

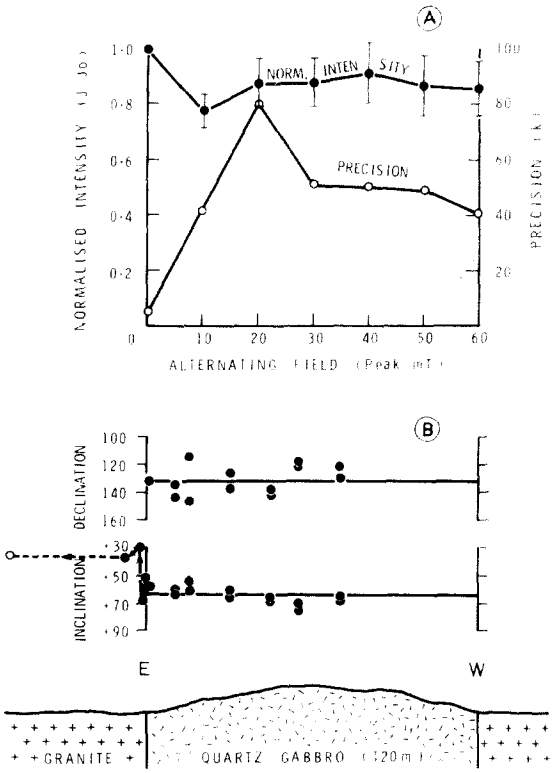


Fig. 3. A, The demagnetization-intensity curve and precision (*k*, Fisher, 1953) plotted as functions of alternating field for pilot specimens from the Cooglegong Creek section in the Black Range dyke.
B, The variation of declination and inclination of magnetization in the dyke with distance from the dyke/granite contact. The dyke's section and topography are shown schematically.

produced by cleaning are shown in Fig. 2C. Magnetic cleaning of all remaining specimens in the optimum field of 20 mT again produced a tight cluster of directions: the core means are shown in Fig. 2. Results from six cores have been rejected from the analysis of directions. Three core mean directions, indicated in Fig. 2E, remained oblique to the principal population and for the other three cores, the resultant, R, calculated giving unit weight to specimens comprising each core, was $< Ro$, the significance point for R at the 95% probability level (Vincenz and Bruckshaw, 1960). Neither the declination nor the inclination displayed any systematic variation across the intrusive (see Fig. 4B). Approximately a quarter of the distance in from the eastern contact, the declination showed a large amount of scatter. The six rejected cores were drilled in the central part of the dyke.

A summary of the site-mean palaeomagnetic data for the Cooglegong Creek and Hillside sections through the Black Range dyke is given in Table I.

By inspection, it can be seen that the declinations measured in the Shaw River section at Hillside are more easterly than the declinations measured at Cooglegong Creek. The application of an F-ratio test to investigate the statistical significance of the difference is inappropriate since the precision parameters (k) are too dissimilar. The small angular difference between the two mean directions probably reflects some large-scale structural deformation imposed since the acquisition of the primary magnetization. The best estimate

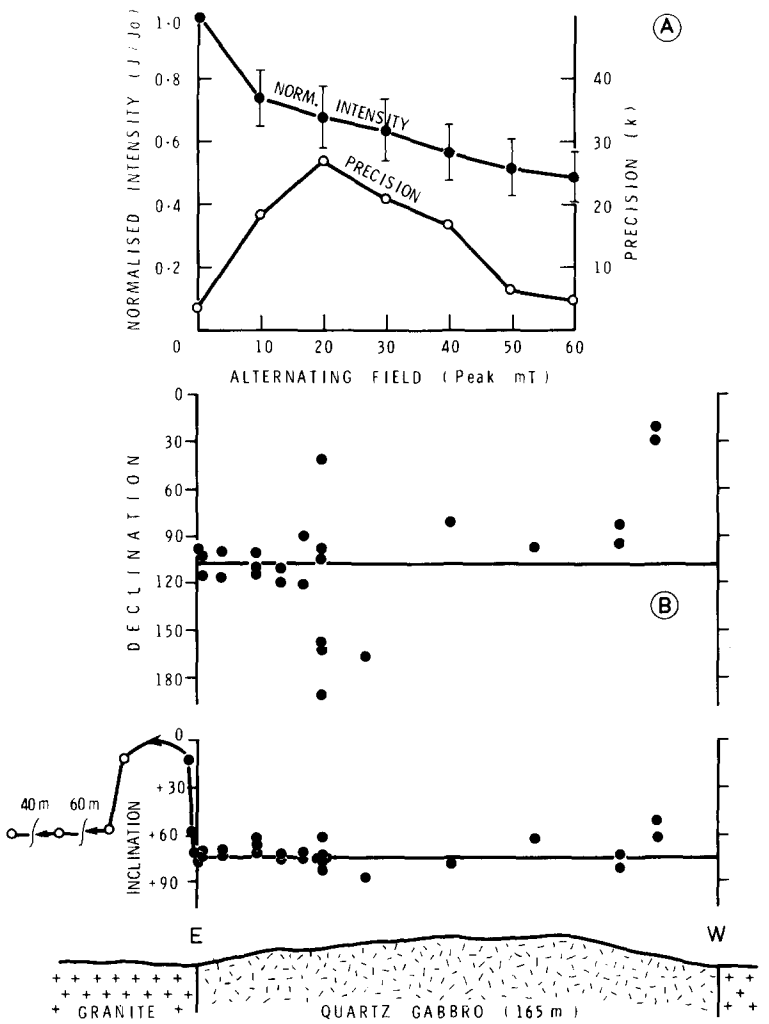


Fig. 4. A, The demagnetization-intensity curve and precision (k, Fisher, 1953) plotted as functions of alternating field for pilot specimens from the Shaw River (Hillside) section in the Black Range dyke.
B, The variation of declination and inclination of magnetization in the dyke with distance from the dyke/granite contact. The dyke's section and topography are shown schematically.

TABLE I

Site-mean directions of remanent magnetization after magnetic cleaning at 20 mT AF for the Black Range dyke

Samples	N*	R	Dec.°	Inc.°	α_{95}°	Estimated distance from eastern margin (m)
Cooglegong Creek						
PD2-6	2	1.995	132.0	+57.5	17.0	½
PD2-8/9	5	4.974	139.5	+62.0	6.0	10
PD2-10/11	5	4.888	125.5	+60.0	13.0	16
PD2-12/13	5	4.979	130.5	+63.5	5.5	30
PD2-14/15	5	4.997	140.0	+68.0	2.0	45
PD2-16/17	5	4.978	119.5	+73.5	5.5	55
PD2-18/19	7	6.897	128.0	+68.0	8.0	70
PD2 Combined	7	6.965	131.0	+65.0	4.5	(k = 170)
Hillside						
PD3-8/9/10	5	4.993	106.5	+74.0	3.0	1
PD3-11/12	4	3.995	108.0	+73.0	4.0	8
PD3-13/14/15	7	6.934	107.0	+67.0	6.5	19
PD3-16/17	4	3.996	115.5	+73.5	3.5	27
PD3-18/19	5	4.968	106.5	+74.5	7.0	34
PD3-20/21/22 23/25/26	13	12.437	128.0	+79.5	9.0	40
PD3-27/30/31	8	7.789	096.0	+79.5	9.5	74
PD3-35/36	4	3.926	091.5	+78.5	14.5	135
PD-37/38	6	5.861	024.5	+57.0	11.5	147
PD3 Combined	9	8.820	093.0	+76.0	8.0	(k = 44)
Mean of site-means	16	15.626	115.0	+72.0	6.0	(k = 40)
Mean of site VGPs						
	Lat. 32°S, Long. 154°E, A_{95} = 9°					

*N is the number of unit vectors combined to produce a mean vector of length R. Dec. = declination and Inc. = inclination. α_{95} (A_{95}) is the semi-angle of the cone of confidence about the true mean direction (mean pole position) calculated at the 95% probability level (Fisher, 1953). VGP is a virtual geomagnetic pole (e.g., see McElhinny, 1973, p. 26).

of the mean direction of magnetization for the Black Range dyke is obtained by averaging the 16 site-means (dec. = 115°, inc. = 72°, α_{95} = 6°) and the mean palaeomagnetic pole position computed from site VGPs lies at 32°S, 154°E, A_{95} = 9°.

The Cajuput dyke

About one-third of the NRM directions grouped in the southeastern

quadrant of the stereogram, and the remaining two-thirds of the NRM core-mean directions appeared to be widely scattered, though mainly on the lower hemisphere as shown in Fig. 5A. The average intensity of magnetization was 20 $mA.m^{-1}$. One specimen from each core was chosen for stepwise/partial AF demagnetization treatment at 10 mT steps to a maximum of 60 mT. The coercive force spectrum analysed in this way is shown in Fig. 6A. Although

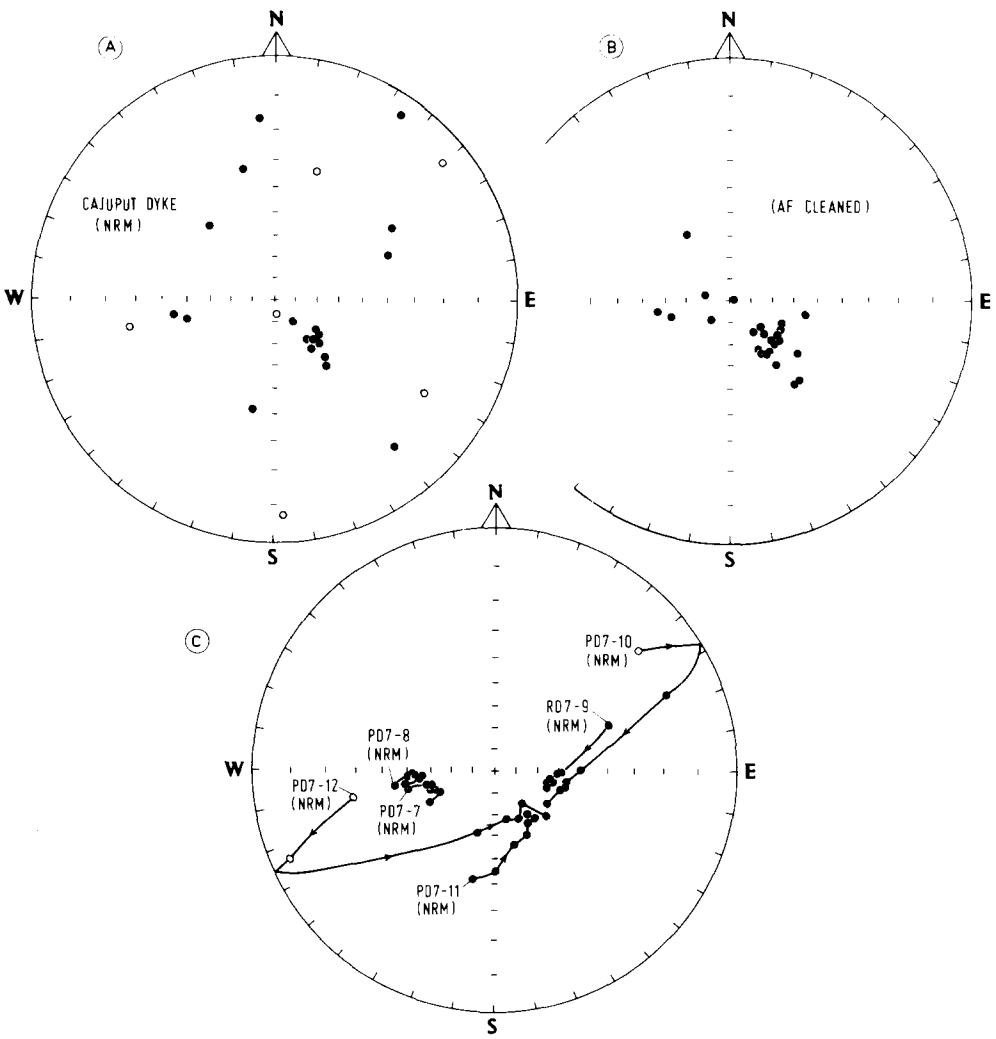


Fig. 5. Core-mean directions of magnetization for the Cajuput dyke shown before and after AF treatment (A and B respectively). Part C shows some examples of direction trends obtained during partial AF demagnetization at 10 mT field increments to 60 mT. Notation as for Fig. 2.

k maintained a low value, it increased systematically to peak after treatment of 50 mT.

Six examples of direction trends in response to magnetic cleaning are shown in Fig. 5C. The overall tendency was for the oblique magnetization vectors to rotate towards the downward pointing tight group of NRM directions in the south-east quadrant. After cleaning all specimens at the optimum

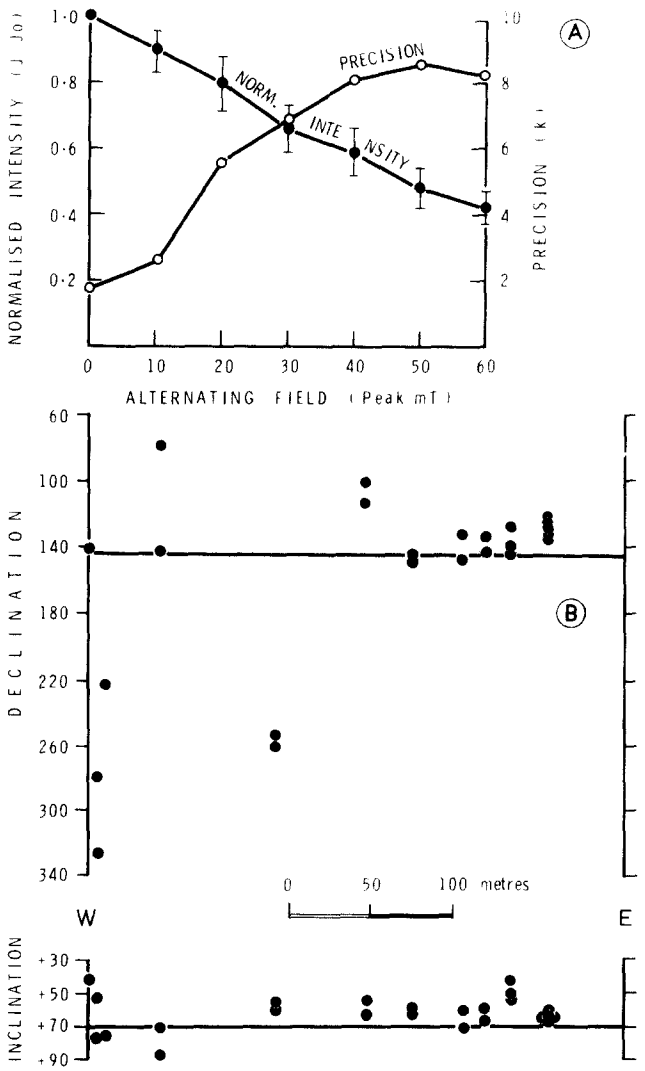


Fig. 6. A, The demagnetization-intensity curve and precision (*k*, Fisher, 1953) plotted as functions of alternating field for pilot specimens from the Cajuput dyke.
B, The variation of declination and inclination of magnetization in the dyke with distance from the dyke/granite contact.

alternating field, the results from only one core, PD7-25, were rejected. The directions of magnetization in the two specimens cut from that core remained separated by 77°. The site-mean data are summarized in Table II and core-mean directions are plotted in Fig. 5B.

Declination and inclination values are plotted as a function of distance from the contact margins in Fig. 6B. Over the western half of the dyke declinations show a large variation, contributing significantly to the rather low precision of the computed result. The mean of the site-mean directions is 145°, + 71° (α_{95} = 13°) and the site mean VGPs yield a pole position at 46° S, 146° E, A_{95} = 22°.

TABLE II

Site-mean directions of remanent magnetization after magnetic cleaning at 50 mT AF for the Cajuput dyke

Samples	N*	R	Dec.°	Inc.°	α_{95}°	Estimated distance from western margin (m)
PD7-1/2/3/4	8	6.865	232.5	+84.5	24.0	10
PD7-5/6	3	2.963	135.5	+83.5	17.0	44
PD7-7/8	4	3.993	257.5	+58.0	4.5	114
PD7-9/10	6	5.957	106.0	+60.0	6.0	169
PD7-11/12	5	4.991	146.5	+61.5	3.5	197
PD7-13/14	6	5.954	141.0	+66.0	6.5	228
PD7-15/16	6	5.944	139.0	+64.0	7.0	242
PD7-17/18/19	6	5.961	140.0	+50.0	6.0	257
PD7-20/21/22 23/24	12	11.986	128.0	+63.0	1.5	280
Mean of site-means:	9	8.489	145	+71	13	(k = 16)
Mean of site VGPs: Lat. 46° S, Long. 146° E, A_{95} = 22°						

*For explanation of terms see Table I.

The baked contact rock

NRM directions measured in samples of the baked granite appear to have been totally reset only within a narrow margin of the actual contact. At Cooglegong Creek, sample PD2-1, collected about 0.3 m from the contact, yielded an inclination of 53° and for PD2-2, 1 m from the contact, the inclination was 69°. PD2-3, collected about 2.5 m from the contact, had an inclination of 22°. Similarly in the Shaw River section at Hillside, PD3-1, 0.6 m from the contact and PD3-2, 1.6 m from the contact had NRM inclinations of 74° and 59°, respectively. At 3.6 m from the contact, sample

PD3-3 yielded an inclination of 16°. Partially reset magnetizations are evident up to 30 m from the contact. These results are summarized for the respective sections through the dyke in Fig. 3B and Fig. 4B.

Partial AF demagnetization revealed a wide spectrum of coercive forces. However, this variation in response to AF treatment with distance from the contact zone was unsystematic and probably reflects the complex nature of the granite's magnetization. With increasing AF treatment, all directions of magnetization rotated towards the NRM directions measured at distances > 30 m from the contact, close to the direction of the present field at the sampling locality.

It is interesting to compare these results with those obtained by Lewis et al. (1974) from a geochemical study of the contact aureole in the Cooglegong Creek section. They concluded that granitic country rock up to 2 m from the contact was remelted and that within the metamorphic aureole, microcline was converted to perthitic orthoclase up to 55 m from the contact. Within about 10 m of the contact, the granite behaved as a completely open system with respect to radiogenic Sr, whilst samples more than 100 m away appear to have been unaffected.

In the palaeomagnetic investigation of the Widgiemooltha dykes, Evans (1968) obtained results from one site in the baked granite about 5 m from the contact. Those data were indistinguishable from results obtained from sites in the dyke. Although the remagnetization of the country rock into which the Black Range dyke was intruded is less widespread, the results when interpreted in terms of the baked contact test in palaeomagnetism are no less significant.

DISCUSSION

The pole position obtained from the Black Range dyke is the first dated pole for the Pilbara craton. According to the classification system of McElhinny and Embleton (1976), the Black Range dyke pole qualifies for key pole status and the Cajuput dyke pole (due to the absence at this stage of radiometric age information) for subsidiary pole status. All of the palaeomagnetic data for the Pilbara and Yilgarn cratons for the period pre-2000 m.y. are shown in Fig. 7. The pole designated YA? represents a group of Yilgarn dykes (Giddings, 1976) whose age of intrusion is uncertain. The Rb-Sr isotopic data are consistent with ages ca. 2500 m.y. or ca. 1700 m.y.: the pole has been included for completeness. Similarly, the pole designated MG3? obtained from the Mt. Goldsworthy iron ore, may equally be much younger than suggested by McElhinny and Embleton (1976): the rather broad age constraints then assigned to pole MG3 (viz., 3000—2000 m.y.) were based on the rather tenuous evidence of lode ore cobbles being found in the Mt. Bruce Supergroup (Brandt, 1964). Although a source lode ore was in existence prior to the estimated minimum age (ca. 1950 m.y.) of the Mt. Bruce Supergroup, it

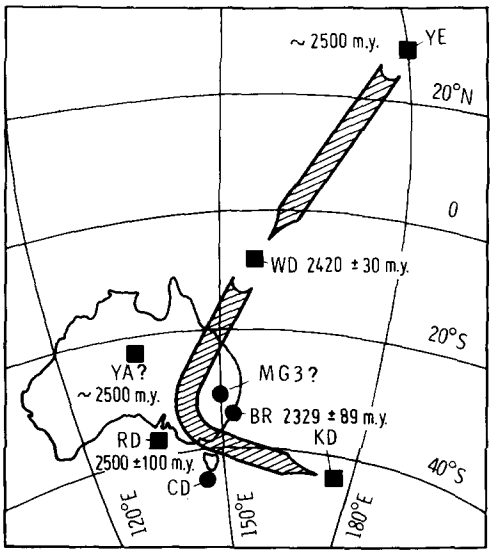


Fig. 7. The apparent polar wander curve for Australia drawn from McElhinny and Embleton (1976) for the period pre-2000 to ca. 2500 m.y. The pole positions for the Black Range dyke (BR) and Cajuput dyke (CD) fall near the curve without violating the pre-existing age constraints. Poles YE, YA? (see text) and RD were obtained by Giddings (1976) from studies on groups of basic dykes from the Yilgarn craton; pole WD is for the Widgiemooltha Dyke Suite in the Yilgarn craton (Evans, 1968). Poles MG3 (3000—2000 m.y.?) and KD (2750—2200 m.y.) are for the Mount Goldsworthy (Pilbara) and the Koolyanobbing-Dowd's Hill (Yilgarn) iron ore deposits (Porath and Chamalaun, 1968) — see text.

is not clear that the Mt. Goldsworthy deposit was that source.

A single apparent polar wander curve can be drawn through the pole positions without violating the chronological constraints. Following the simplest interpretation then, the palaeomagnetic data from the Pilbara and Yilgarn cratons, for the interval of time about 2400 m.y., are consistent with an ensialic origin for the younger mobile belt which separates the two cratons. Although relative movement between the cratons remains a possibility according to the alternative models described in the Introduction, the possibility is receding as more results are reported (e.g., McElhinny and McWilliams, 1977).

Plate tectonic models evolved from studies of ocean basins may not be appropriate to describe structural evolution in the Proterozoic. It may also be that the number of plates has increased since Precambrian time so that during the Precambrian, the probability of plate convergence was diminished (McElhinny and McWilliams, 1977). One cannot dispute the long history of volcanism experienced during the Precambrian, e.g., the almost ubiquitous occurrence of greenstone belts within the shield regions, but the 'island arc' type of environment would seem to have been far less common than it is today.

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